Summary 1

Systems of Linear Equations

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1.1 Systems of Linear Equations

1.1.1 Forming the Equations

$$\begin{aligned} a+b+c&=176 \\ 0.85a+0.90b+0.95c&=161.80 \\ 0.93a+0.95b+0.98c&=169.56 \end{aligned}$$

Using GC,

$$\begin{aligned} a&=32 \\ b&=44 \\ c&=100 \end{aligned}$$

The "variables" may be expressions. To find x, y and z, use the given information to form the equations:

$$2x^3+3\ln y-\mathrm{e}^z=18, \qquad x^3-\ln y+2\mathrm{e}^z=9, \qquad x^3+2\ln y+\mathrm{e}^z=11$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=\mathrm{e}^z$, which turns it into a linear system in A, B and C. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because x appears as both x^2 and x, and there are product terms such as xz and xy.

1.1.2 Special Case: Number of Equations > Number of Variables

Sometimes there may be no solution, e.g. skew lines in vectors.

$$\begin{aligned} \lambda+\mu&=1 \\ 2\lambda-\mu&=5 \\ \lambda+3\mu&=-1 \end{aligned}$$

Using GC,

$$\begin{aligned} \lambda&=2 \\ \mu&=-1 \end{aligned}$$

1.1.3 Special Case: Number of Equations < Number of Variables

$$\begin{aligned} x+y+z&=6 \\ x-y+2z&=5 \end{aligned}$$

Using GC,

$$\begin{aligned} x&=\tfrac{11}{2}-\tfrac{3}{2}z \\ y&=\tfrac{1}{2}+\tfrac{1}{2}z \\ z&=z \end{aligned}$$

Note

Useful trick: in contextual questions, the variables are usually non-negative integers, $z\in\mathbb{Z}_0^+$, so use this to find the solution from the general solution.

Summary 2

Inequalities

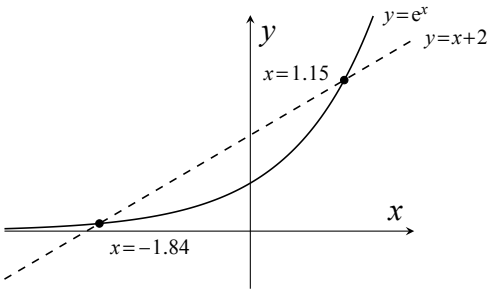
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2.1 Solving Inequalities Graphically

- Rearrange to $\mathrm{f}(x) > 0$, or compare $y = \mathrm{f}(x)$ with $y = \mathrm{g}(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the x-values where the required graph is above or below.

$$\mathrm{e}^x < x + 2$$

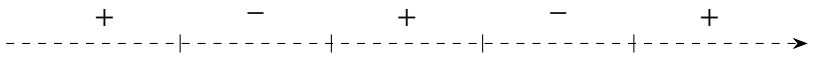


Using GC, intersections at $x = -1.8414$, $x = 1.1462$

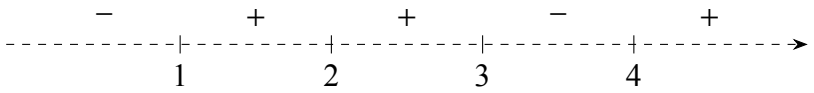
$$\therefore\ -1.84 < x < 1.15$$

2.2 Sign Test Technique

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant <0 and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.



$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$



$x=2$: numerator $=0$ (included)

$x=3,\ x=4$: denominator $=0$ (excluded)

$\therefore x \leqslant 1$ or $x=2$ or $3<x<4$

> **Note**
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the same.

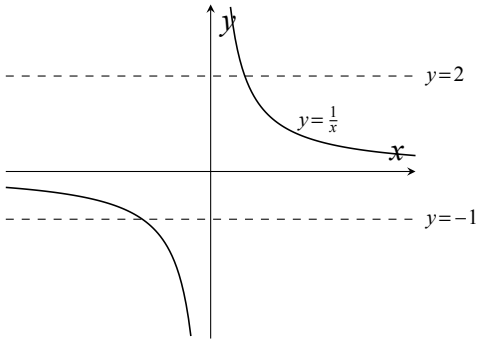
2.3 Common Substitutions

Replace x by	Original inequality		
	$x>2$	$x<-1$	$-1<x<2$
e^x	$x>\ln 2$	no solution	$x<\ln 2$
$\ln x$	$x>e^2$	$0<x<e^{-1}$	$e^{-1}<x<e^2$
$-x$	$x<-2$	$x>1$	$-2<x<1$
$x+1$	$x>1$	$x<-2$	$-2<x<1$
$\lvert x\rvert$	$x<-2$ or $x>2$	no solution	$-2<x<2$
x^2	$x<-\sqrt{2}$ or $x>\sqrt{2}$	no solution	$-\sqrt{2}<x<\sqrt{2}$
$\frac{1}{x}$ **(VERY IMPORTANT)**	see graph	see graph	see graph
$\cos x$	see graph	see graph	see graph

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

2.3.1 Example: $x \to \frac{1}{x}$

$$\frac{1}{x}>2, \qquad \frac{1}{x}<-1, \qquad -1<\frac{1}{x}<2$$



2.3.2 Example: $x \to \cos x$

$\cos x > \frac{1}{2}$, $\cos x < -\frac{1}{2}$, $-\frac{1}{2} < \cos x < \frac{1}{2}$, $0 \leqslant x \leqslant 2\pi$

